# Worksheet 5.1 — Optimization

Grade 12 Calculus and Vectors (MCV4U) · Unit 5: Applications of Derivatives

## The Big Ideas

Write the quantity to optimize, reduce it to one variable using the constraint, solve $Q' = 0$, and verify.

- Define a variable; write $Q$.
- Use the constraint to make $Q$ one-variable; state the domain.
- Solve $Q' = 0$; verify with the 1st/2nd derivative test.

## The Big Picture

### Best possible value

**Optimization** finds a maximum or minimum in a real situation  largest area, least cost, shortest time.

### The strategy

Write the quantity to optimize as a function of one variable (using any constraint), then find its critical points on the valid domain.

### Confirm the extreme

Use the first or second derivative test, and check the endpoints, to confirm you found the true maximum or minimum.

## Worked Examples

### Example 1 — Max area, fixed perimeter

A pen uses 40 m of fence. Maximize the area.

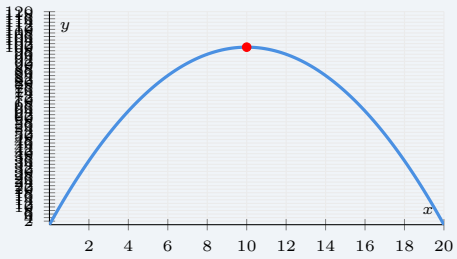
***Solution.*** **Step 1 (define a variable and write the constraint):** let the width be $x$; since $2x+2y = 40$, the length is $y = 20 - x$.

**Step 2 (write the quantity to optimize, in one variable):** $A(x) = x(20 - x)$.

**Step 3 (differentiate and solve $A' = 0$):** $A'(x) = 20 - 2x = 0 \Rightarrow x = 10$.

**Step 4 (verify it's a max):** $A''(x) = -2 < 0$, so concave down — a maximum.

**Conclusion:** a $10 \times 10$ square, giving maximum area 100 m$^2$:



## Example 2 — Wall on one side

Only 3 sides fenced with 20 m. Maximize the area.

***Solution.*** **Step 1 (define a variable and the constraint):** let $x$ be each of the two equal sides perpendicular to the wall; the constraint is $2x + y = 20$, so $y = 20 - 2x$.

**Step 2 (write $A$ in one variable):** $A(x) = x(20 - 2x)$.

**Step 3 (differentiate and solve $A' = 0$):** $A'(x) = 20 - 4x = 0 \Rightarrow x = 5$.

**Step 4 (find the other dimension and the area):** $y = 20 - 10 = 10$; $A = 5(10) = 50$.

**Conclusion:** maximum area 50 $\text{m}^2$, with the fenced sides 5 m and 10 m.

## Example 3 — Minimize a sum

For $x > 0$, minimize $f(x) = x + \dfrac{16}{x}$.

***Solution.*** **Step 1 (differentiate):** $f'(x) = 1 - \dfrac{16}{x^2}$.

**Step 2 (set $f' = 0$ and solve):** $1 = \dfrac{16}{x^2} \Rightarrow x^2 = 16 \Rightarrow x = 4$ (since $x > 0$).

**Step 3 (verify it's a minimum):** $f''(x) = \dfrac{32}{x^3} > 0$ for $x > 0$, so concave up — a minimum.

**Step 4 (find the minimum value):** $f(4) = 4 + \dfrac{16}{4} = 4 + 4$.

**Conclusion:** minimum value 8, at $x = 4$.

## Example 4 — Max-volume open box

Cut side-$x$ squares from a $12 \times 12$ sheet; fold into an open box. Maximize the volume.

***Solution.*** **Step 1 (set up the dimensions):** folding up flaps of size $x$ leaves a base of side $12 - 2x$ and height $x$.

**Step 2 (write $V$ in one variable):** $V(x) = x(12 - 2x)^2$.

**Step 3 (differentiate using the product rule with the chain rule):** $V'(x) = (12 - 2x)^2 + x \cdot 2(12 - 2x)(-2) = (12 - 2x)[(12 - 2x) - 4x] = (12 - 2x)(12 - 6x)$.

**Step 4 (solve $V' = 0$ within the valid domain $0 < x < 6$):** $12 - 6x = 0 \Rightarrow x = 2$ (the factor $12 - 2x = 0$ gives $x = 6$, an endpoint with zero volume).

**Step 5 (compute the volume):** $V(2) = 2(8)^2 = 2(64)$.

**Conclusion:** maximum volume 128 $\text{cm}^3$, at $x = 2$.

## Example 5 — Maximum profit

Maximize $P(x) = -2x^2 + 40x - 100$.

***Solution.*** **Step 1 (differentiate):** $P'(x) = -4x + 40$.

**Step 2 (set $P' = 0$ and solve):** $-4x + 40 = 0 \Rightarrow x = 10$.

**Step 3 (verify it's a max):** $P''(x) = -4 < 0$, so concave down — a maximum.

**Step 4 (find the maximum value):** $P(10) = -2(100) + 400 - 100 = -200 + 400 - 100$.

**Conclusion:** maximum profit 100, at $x = 10$.

### Example 6 — Max product

Two positive numbers sum to 12. Maximize the product.

***Solution.*** **Step 1 (define variables using the constraint):** let the numbers be $x$ and $12 - x$.

**Step 2 (write the product):** $P(x) = x(12 - x)$.

**Step 3 (differentiate and solve $P' = 0$):** $P'(x) = 12 - 2x = 0 \Rightarrow x = 6$.

**Step 4 (find the product):** both numbers are 6, so $P = 6(6)$.

**Conclusion:** maximum product 36, when both numbers equal 6.

### Example 7 — Max area

Maximize the area of a rectangle with 60 m of fence.

***Solution.*** **Step 1 (define a variable and the constraint):** let the width be $x$; since $2x + 2y = 60$, $y = 30 - x$.

**Step 2 (write $A$ in one variable and optimize):** $A(x) = x(30 - x)$, $A'(x) = 30 - 2x = 0 \Rightarrow x = 15$.

**Step 3 (find the dimensions and area):** $y = 30 - 15 = 15$; $A = 15(15)$.

**Conclusion:** a $15 \times 15$ square, giving maximum area 225 $\text{m}^2$.

### Example 8 — Minimize a sum

For $x > 0$, minimize $f(x) = x + \dfrac{9}{x}$.

***Solution.*** **Step 1 (differentiate):** $f'(x) = 1 - \dfrac{9}{x^2}$.

**Step 2 (set $f' = 0$ and solve):** $x^2 = 9 \Rightarrow x = 3$ (since $x > 0$).

**Step 3 (find the minimum value):** $f(3) = 3 + \dfrac{9}{3} = 3 + 3$.

**Conclusion:** minimum value 6, at $x = 3$.

### Example 9 — Maximum value

Maximize $P(x) = -x^2 + 10x$.

***Solution.*** **Step 1 (differentiate):** $P'(x) = -2x + 10$.

**Step 2 (set $P' = 0$ and solve):** $-2x + 10 = 0 \Rightarrow x = 5$.

**Step 3 (find the maximum value):** $P(5) = -25 + 50$.

**Conclusion:** maximum value 25, at $x = 5$.

## Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

### Question 1

Two positive numbers sum to 20. Maximize the product.

### Question 2

Maximize the area of a rectangle with 80 m of fence.

### Question 3

For $x > 0$, minimize $f(x) = x + \dfrac{25}{x}$.

### Question 4

Maximize $P(x) = -x^2 + 8x$.

### Question 5

A 3-sided pen uses 40 m of fence. Maximize the area.

## Question 6

Two positive numbers sum to 30. Maximize the product.

## Question 7

For $x > 0$, minimize $f(x) = x + \dfrac{4}{x}$.

## Question 8

Maximize $P(x) = -3x^2 + 30x$.

## Question 9

Cut side-$x$ squares from a $6 \times 6$ sheet for an open box. Find $x$ that maximizes the volume.

## Question 10

Maximize the area of a rectangle with 100 m of fence.

## Question 11

For $x > 0$, minimize $f(x) = x + \dfrac{1}{x}$.

## Question 12

Maximize $P(x) = -2x^2 + 12x - 5$.

## Question 13 — Challenge

A farmer has 240 m of fence to make two identical adjacent rectangular pens (so three lengths of fence run one way). Maximize the total area.

# Question Recap & Answer Key

## Question Recap

1. Two positive numbers sum to 20. Maximize the product.
2. Maximize the area of a rectangle with 80 m of fence.
3. For $x > 0$, minimize $f(x) = x + \dfrac{25}{x}$.
4. Maximize $P(x) = -x^2 + 8x$.
5. A 3-sided pen uses 40 m of fence. Maximize the area.
6. Two positive numbers sum to 30. Maximize the product.
7. For $x > 0$, minimize $f(x) = x + \dfrac{4}{x}$.
8. Maximize $P(x) = -3x^2 + 30x$.
9. Cut side-$x$ squares from a $6 \times 6$ sheet for an open box. Find $x$ that maximizes the volume.
10. Maximize the area of a rectangle with 100 m of fence.
11. For $x > 0$, minimize $f(x) = x + \dfrac{1}{x}$.
12. Maximize $P(x) = -2x^2 + 12x - 5$.
13. A farmer has 240 m of fence to make two identical adjacent rectangular pens (so three lengths of fence run one way). Maximize the total area.

## ✓ Answer Key

1. $10, 10$; product 100
2. $20 \times 20$, 400 mš
3. $x = 5$; min 10
4. $x = 4$; $P = 16$
5. $10 \times 20$, 200 mš
6. $15, 15$; product 225
7. $x = 2$; min 4
8. $x = 5$; $P = 75$
9. $x = 1$; $V = 16$
10. $25 \times 25$, 625 mš
11. $x = 1$; min 2
12. $x = 3$; $P = 13$
13. width $x$, $3x + 2L = 240$; $A = x \cdot \frac{240-3x}{2}$, $x = 40$, $A = 2400$ mš